SMALL UNIVALENT CATEGORIES AND FUNCTORS OF POINTS REPLACEMENT, FINITE PRESENTATIONS, AND QCQS SCHEMES

Abstract. A category can be presented as a 1-type with sets of arrows, identities, and composition, with its identity types canonically equivalent to its sets of isomorphisms. We show how replacement produces the small full category generated by a small family in a locally small category. Its universal property follows from descent along fully faithful, essentially surjective functors. Applied to finite presentations, this constructs a small univalent category whose objects are precisely the finitely presented commutative algebras over a fixed ring. Yoneda then gives the fully faithful contravariant embedding into set-valued functors. We then prove that full faithfulness and mere essential surjectivity between univalent categories give an explicit categorical equivalence without choice. Applied to qcqs schemes, this identifies the geometric category of locally ringed lattices with the category of functors satisfying the Zariski sheaf and finite affine atlas conditions, both absolutely and over an arbitrary small base ring. The last section records the correspondence with the checked implementation.

1. Categories as structured 1-types

We work in univalent type theory with a hierarchy of universes $\mathcal{U}_\ell$. A proposition is a type any two of whose elements are equal; a set is a type whose identity types are propositions; a 1-type is a type whose identity types are sets. We write $\|T\|$ for propositional truncation, so that $\|T\|$ expresses mere existence of an element of T. Function extensionality follows from univalence.

A *precategory* C consists of a type $\mathrm{Ob}(C)$, sets $C(x,y)$, identities 1_x, and composition

$$C(y,z) \times C(x,y) \longrightarrow C(x,z), \qquad (g,f) \longmapsto g \circ f,$$

satisfying the two unit laws and associativity. Its objects need not form a set. For an arrow f, the type of inverse arrows together with the two inverse equations is a proposition: the inverse is unique, and the inverse equations are propositions. Consequently $\mathrm{Iso}_C(x,y)$ is a set.

Path induction gives a canonical function

$$\mathrm{idtoiso}_{x,y} : (x = y) \longrightarrow \mathrm{Iso}_C(x,y), \qquad \mathrm{idtoiso}(\mathrm{refl}_x) = 1_x.$$

Definition 1.1. A *category* is a precategory for which every function $\mathrm{idtoiso}_{x,y}$ is an equivalence. Its inverse is denoted isotoid.

It follows that the objects of a category form a 1-type. Thus we can regard a category as a 1-type equipped with hom sets and category operations, subject also to the requirement that the canonical maps $(x = y) \to \mathrm{Iso}_C(x,y)$ be equivalences. The 1-type condition alone is insufficient. All arrows remain part of the structure, including the noninvertible ones. The equality $(x = x) \simeq \mathrm{Aut}_C(x)$ explains why this object type is not generally a set of isomorphism classes.

Date: October 29, 2026.

The following criterion lets us prove univalence by a uniqueness argument.

Lemma 1.2 (Total-space criterion)**.** *A precategory C is a category if and only if, for every x, the type*

$$S_x = \sum_{y:\mathrm{Ob}(C)} \mathrm{Iso}_C(x, y)$$

is a proposition. In that case S_x is contractible, with centre $(x, 1_x)$.

Proof. In a category, S_x is equivalent to $\sum_y (x = y)$, which is contractible. Conversely, suppose S_x is a proposition. For $e : x \cong y$, project the unique equality $(x, 1_x) = (y, e)$ onto the object coordinate to obtain $p_e : x = y$. Path induction on this equality in S_x gives $\mathrm{idtoiso}(p_e) = e$. Path induction on $p : x = y$ gives $p_{\mathrm{idtoiso}(p)} = p$: in the reflexive case the equality $(x, 1_x) = (x, 1_x)$ is reflexive since S_x is a proposition. These are the required inverse laws. □

A functor is *fully faithful* if each of its functions on hom sets is an equivalence. It is *essentially surjective* if

$$\prod_{b:\mathrm{Ob}(B)} \left\| \sum_{c:\mathrm{Ob}(C)} \mathrm{Iso}_B(qc, b) \right\|$$

is inhabited. We distinguish this from *split* essential surjectivity, which supplies an actual pair (c, e) for each b. Full faithfulness reflects isomorphisms: lift an inverse by fullness and check its two equations by faithfulness.

2. Functor categories

For precategories B, E, functors $B \to E$ and natural transformations form a precategory $\mathrm{Fun}(B, E)$. A natural transformation $\alpha : F \to G$ is a family of arrows $\alpha_b : Fb \to Gb$ such that

$$G(f) \circ \alpha_b = \alpha_{b'} \circ F(f) \qquad (f : b \to b').$$

Naturality is a proposition. Transformations form a set, and are equal when their components are equal. Their identities and composition are componentwise.

Proposition 2.1. *If E is a category and B is any precategory, then* $\mathrm{Fun}(B, E)$ *is a category. In particular,*

$$(F = G) \simeq \mathrm{Iso}_{\mathrm{Fun}(B,E)}(F, G).$$

Proof. An isomorphism of functors has isomorphisms as its components. Conversely, if all components of a natural transformation are isomorphisms, their inverses are natural by multiplying the naturality square by the two inverses. They give its inverse transformation.

Fix F. Consider the total type of pairs (G, α) where $\alpha : F \cong G$. Its object data are a family $G_0(b)$ and isomorphisms $a_b : Fb \cong G_0(b)$. The type of such data is equivalent to

$$\prod_{b:\mathrm{Ob}(B)} \left(\sum_{y:\mathrm{Ob}(E)} \mathrm{Iso}_E(Fb, y) \right),$$

which is contractible by Lemma 1.2. For fixed object data, naturality forces the arrow action to be

$$G_1(f) = a_{b'} \circ F(f) \circ a_b^{-1}.$$

Thus there is at most one arrow action. The functor laws and naturality are propositions. The total type (G, α) is therefore a proposition, and Lemma 1.2 applies to $\mathrm{Fun}(B, E)$. □

3. Replacement and the small full image

A type is $\mathcal{U}_\ell$-*locally small* if each of its identity types is equivalent to a type in $\mathcal{U}_\ell$. We use the following form of replacement. For $P : \mathcal{U}_\ell$, a locally small type $T : \mathcal{U}_k$, and a function $F : P \to T$, there are

$$P \xrightarrow{q} I \xrightarrow{i} T, \qquad I : \mathcal{U}_\ell, \qquad i(qp) = F(p),$$

where i is an embedding and q is surjective. Here being an embedding means that each function

$$(a = b) \longrightarrow (i(a) = i(b))$$

is an equivalence. Surjectivity permits elimination into dependent families of propositions in any universe. The small image I is the same for all these elimination universes. This last clause matters when membership in the image is a proposition in a larger universe.

Theorem 3.1 (Small full image)**.** *Let D be a category with hom sets in $\mathcal{U}_\ell$, and let $F : P \to \mathrm{Ob}(D)$ with $P : \mathcal{U}_\ell$. There is a category D_F with both objects and hom sets in $\mathcal{U}_\ell$, together with a fully faithful inclusion $i : D_F \to D$, such that*

$$\mathrm{fib}_i(x) \simeq \left\| \sum_{p:P} \mathrm{Iso}_D(Fp, x) \right\|. \tag{1}$$

The fibre on the left is a proposition.

Proof. The object type of D is locally small because $(x = y) \simeq \mathrm{Iso}_D(x, y)$ and the latter type is small. Apply replacement to F, obtaining $P \to I \to \mathrm{Ob}(D)$. Define

$$\mathrm{Ob}(D_F) = I, \qquad D_F(a, b) = D(ia, ib),$$

with identities and composition inherited from D. The inclusion is the identity on hom sets and is fully faithful. Its univalence follows from

$$(a = b) \simeq (ia = ib) \simeq \mathrm{Iso}_D(ia, ib) = \mathrm{Iso}_{D_F}(a, b).$$

By path induction this composite is the canonical idtoiso.

Since i is an embedding, $\mathrm{fib}_i(x) = \sum_{a:I}(ia = x)$ is a proposition. A pair $(p, e : Fp \cong x)$ determines an element of this fibre by univalence and $i(qp) = Fp$. We can therefore eliminate its mere existence into the fibre. Conversely, a fibre point (a, r) and surjectivity of q give merely some p with $qp = a$, hence an isomorphism $Fp \cong x$. Both sides of (1) are propositions, so these maps give an equivalence. □

There is also a precategory C_F with objects P and arrows $C_F(p, p') = D(Fp, Fp')$. The map q extends to a functor $q : C_F \to D_F$. More explicitly, write $t_p : i(qp) \cong Fp$ for the isomorphism given by the image equation. Then

$$q(h) = t_{p'}^{-1} \circ h \circ t_p.$$

Conjugation is an equivalence on hom sets, and every object of D_F is merely isomorphic to some qp. Thus q is fully faithful and essentially surjective. We next give its universal property.

4. Descent along a fully faithful, essentially surjective functor

Throughout this section let $q : C \to B$ be fully faithful and essentially surjective. The source and target may initially be precategories.

Lemma 4.1. *For any precategory E, precomposition*

$$q^* : \mathrm{Fun}(B, E) \longrightarrow \mathrm{Fun}(C, E)$$

is fully faithful.

Proof. Let $G, H : B \to E$ and $\beta : Gq \to Hq$ be natural. For each b, seek an arrow $\gamma : Gb \to Hb$ satisfying, for every c and every arrow $u : qc \to b$,

$$\gamma \circ G(u) = H(u) \circ \beta_c. \tag{2}$$

The type T_b of an arrow with these equations is a proposition. Indeed, to compare two arrows in it, we may merely choose $e : qc \cong b$, since their equality is a proposition. Equation (2) at e and cancellation of the isomorphism $G(e)$ give equality; the equations attached to the arrows are propositions.

To inhabit T_b, we can now eliminate essential surjectivity into this proposition. For a given $e_0 : qc_0 \cong b$, put

$$\gamma_0 = H(e_0) \circ \beta_{c_0} \circ G(e_0)^{-1}.$$

For any $u : qc \to b$, full faithfulness gives the unique $f : c \to c_0$ with $qf = e_0^{-1}u$. Naturality of β proves (2) for $u = e_0 \circ qf$. Hence T_b is contractible. Its unique arrow defines α_b.

For $g : b \to b'$, compare $H(g)\alpha_b$ and $\alpha_{b'}G(g)$ after composition with $G(e)$ for a merely chosen $e : qc \cong b$. Their equality follows from (2) at e and at ge. Cancellation gives naturality. At $b = qc$, taking $u = 1_{qc}$ gives $\alpha_{qc} = \beta_c$. Finally, any natural transformation restricting to β satisfies (2), so it equals α. This constructs an inverse to precomposition on each hom set. □

Lemma 4.2. *If E is a category, every functor $G : C \to E$ has a descent $H : B \to E$ with a natural isomorphism $Hq \cong G$.*

Proof. We construct Hb without choosing a representative of b. Define X_b to consist of an object $x : E$ and isomorphisms

$$\eta_{c,e} : Gc \cong x \qquad (c : C,\ e : qc \cong b)$$

satisfying

$$\eta_{c',e'} \circ G(f) = \eta_{c,e} \quad \text{whenever } e' \circ qf = e. \tag{3}$$

We first prove X_b is a proposition. This assertion is itself a proposition, so essential surjectivity lets us suppose $e_0 : qc_0 \cong b$ is given. Evaluation at (c_0, e_0) maps X_b to

$$S_{Gc_0} = \sum_{x:E} \mathrm{Iso}_E(Gc_0, x).$$

It has a reconstruction: from (x, φ) define

$$\eta_{c,e} = \varphi \circ G(f_e), \qquad qf_e = e_0^{-1} \circ e.$$

Full faithfulness gives f_e, and reflects its invertibility. Faithfulness and the functor laws give (3). That same equation shows that reconstructing an element of X_b from its evaluation gives the original element. Hence X_b is a retract of S_{Gc_0}, which is a proposition by Lemma 1.2. Thus X_b is a proposition.

Only now do we use essential surjectivity to inhabit X_b: given (c_0, e_0), reconstruct from $(Gc_0, 1)$. We obtain a unique element of X_b, denoted (Hb, η_b).

For $g : b \to b'$, define Y_g to be the type of arrows $\varphi : Hb \to Hb'$ satisfying

$$\varphi \circ \eta_{b,c,e} = \eta_{b',c',e'} \circ G(f) \quad \text{if } e' \circ qf = g \circ e. \tag{4}$$

This is a proposition. To compare two such arrows, merely choose $e : qc \cong b$ and $e' : qc' \cong b'$, lift $e'^{-1}ge$ to f, and cancel $\eta_{b,c,e}$. To inhabit it, use the same representatives and define

$$\varphi_0 = \eta_{b',c',e'} \circ G(f) \circ \eta^{-1}_{b,c,e}.$$

For other representatives $(d, u), (d', u')$, lift $e^{-1}u$ and $e'^{-1}u'$ to $m : d \to c$ and $m' : d' \to c'$. If $u' \circ qh = g \circ u$, faithfulness gives $fm = m'h$. Equation (3) and this equality verify (4) for φ_0. Thus Y_g is contractible; its arrow defines $H(g)$.

The identity satisfies (4) by (3), so $H(1) = 1$. For composable g, g', merely choose a representative for their intermediate object. Lift the two arrows and use faithfulness to identify the composite of the lifts with a lift of $g'g$. Then $H(g')H(g)$ satisfies (4) for $g'g$, so uniqueness gives the composition law. Finally, $\eta_{qc,c,1} : Gc \cong H(qc)$ is natural by (4). Its inverse gives $Hq \cong G$. □

The order of these arguments is essential: the types of possible values are proved to be propositions before a merely existing representative is used to inhabit them. No choice of representatives is required.

Proposition 4.3. *A fully faithful, split essentially surjective functor $P : X \to Y$ between categories induces an equivalence* $\mathrm{Ob}(X) \simeq \mathrm{Ob}(Y)$.

Proof. For $y : Y$, the type $S_y = \sum_{x:X} \mathrm{Iso}_Y(Px, y)$ is a proposition. Indeed, for $(x, e), (x', e')$, lift the isomorphism $e'^{-1}e$ along P to $\varphi : x \cong x'$. Univalence of X makes it a path $x = x'$. Path induction on this path, using $e'P\varphi = e$, identifies the two pairs. Univalence of Y identifies S_y with the fibre $\sum_x (Px = y)$. The splitting inhabits that fibre, so every fibre is contractible. □

Theorem 4.4 (From mere essential surjectivity to equivalence)**.** *Let $P : X \to Y$ be a fully faithful, essentially surjective functor between univalent categories. Then its object map is an equivalence of types, and P has a quasi-inverse functor with natural isomorphisms for both composites. Split essential surjectivity is not an additional hypothesis.*

Proof. Full faithfulness gives equivalences on isomorphism types: lift an isomorphism and its inverse by fullness, and prove the inverse equations by faithfulness. The square

$$\begin{array}{ccc} (x = x') & \xrightarrow{\mathrm{ap}\, P} & (Px = Px') \\ \downarrow\simeq & & \downarrow\simeq \\ \mathrm{Iso}_X(x, x') & \xrightarrow{\simeq} & \mathrm{Iso}_Y(Px, Px') \end{array}$$

commutes by path induction. Its vertical maps are the canonical univalence equivalences, so $\mathrm{ap}\, P$ is an equivalence. Consequently the object map of P is an embedding, and every fibre

$$T_y = \sum_{x:\mathrm{Ob}(X)} (Px = y)$$

is a proposition. Essential surjectivity and univalence of Y give $\|T_y\|$. Eliminate this truncation into the already propositional T_y. Each fibre is now inhabited and contractible, which proves the asserted equivalence on object types.

Write its fibre centres as $(Ky, p_y : PKy = y)$, and let $e_y : PKy \cong y$ be the path-isomorphism of p_y. For $a : y \to z$, define Ka by the unique lift

$$P(Ka) = e_z^{-1} \circ a \circ e_y. \tag{5}$$

Equivalently, Ka is the unique arrow satisfying $e_z \circ P(Ka) = a \circ e_y$. The identity arrow satisfies this square for 1_y, and pasting the squares for two composable arrows proves the square for their composite. Uniqueness proves the two functor laws. The squares themselves

state that e is a natural isomorphism $PK \cong 1_Y$. Reflect e_{Px} along the fully faithful P to obtain $KPx \cong x$. Applying faithfulness to the naturality of e proves naturality of these reflected isomorphisms, giving $KP \cong 1_X$. Thus K is a quasi-inverse. Every truncation was eliminated into a proposition; no axiom of choice is used. □

Theorem 4.5 (Universal property of descent)**.** *For every category E, precomposition by q is fully faithful and split essentially surjective. It induces an equivalence of types*

$$\mathrm{Fun}(B, E) \simeq \mathrm{Fun}(C, E).$$

Proof. Lemmas 4.1 and 4.2 give the two assertions about precomposition. Both functor categories are categories by Proposition 2.1; apply Proposition 4.3. □

Applying this to $C_F \to D_F$ gives

$$\mathrm{Fun}(D_F, E) \simeq \mathrm{Fun}(C_F, E). \tag{6}$$

Thus $q : C_F \to D_F$ has the universal property of the *univalent completion*, also called the *Rezk completion*, of C_F. Here the existing ambient category D and replacement provide a small construction of that completion. Replacement supplies the small object type; the descent argument supplies the universal property.

5. The category of commutative algebras

Fix a commutative unital ring R whose carrier is a set in $\mathcal{U}_\ell$. The zero ring is allowed. An R-algebra is a commutative unital ring A with carrier a set in $\mathcal{U}_\ell$, together with a unital ring map $\iota_A : R \to A$. An R-algebra map $f : A \to B$ is a unital ring map satisfying $f\iota_A = \iota_B$. Such maps form a set, denoted $\mathrm{Hom}_R(A, B)$. Pointwise identity and composition define a precategory Alg_R with objects in $\mathcal{U}_{\ell+1}$ and hom sets in $\mathcal{U}_\ell$.

Lemma 5.1. *An R-algebra map is an isomorphism if and only if its underlying function is an equivalence of sets.*

Proof. An algebraic inverse is a function inverse. Conversely, let g be the inverse function of an algebra map $f : A \to B$. Applying the injective map f to both sides proves

$$g(1) = 1, \qquad g(b + b') = g(b) + g(b'), \qquad g(bb') = g(b)g(b'), \qquad g(\iota_B(r)) = \iota_A(r).$$

For example, both sides of the multiplication equation have image bb'. Thus g is an algebra map, and the function inverse laws are algebra inverse laws. □

Proposition 5.2 (Structure identity principle)**.** *The precategory Alg_R is a category.*

Proof. On a fixed set T, algebra structure consists of the operations $0, 1, +, \cdot, -$, a function $R \to T$, and their laws. The laws are propositions. By universe univalence, equality of carriers is equivalent to equivalence of carriers. Transporting the operations along a carrier equivalence identifies algebra structures exactly when that equivalence preserves the operations and the structure map.

This last assertion follows by equivalence induction: for the identity equivalence, preservation says the constants and operations agree; function extensionality makes these equalities of functions, and the laws are propositions. The equivalence-induction argument is the identity-elimination argument for carriers under universe univalence.

Preservation of $1, +, \cdot$ already implies preservation of 0 and negation, by uniqueness of additive identities and inverses. Thus the structure-preserving equivalences are exactly algebra maps whose carrier maps are equivalences. By Lemma 5.1 they are exactly algebra isomorphisms. This identifies the canonical function $(A = B) \to \mathrm{Iso}_R(A, B)$ as an equivalence. □

6. Finite presentations and a small category of algebras

A finite presentation p consists of a natural number n and finitely many polynomial equations

$$f_1 = g_1, \ldots, f_m = g_m \qquad \text{in } R[X_1, \ldots, X_n].$$

Write $\mathrm{Sol}_p(B)$ for the set of tuples in B^n satisfying these equations. The type $\mathcal{P}_R$ of such presentations belongs to $\mathcal{U}_\ell$. A presentation is *realised* by an algebra A if it has a tuple $a \in \mathrm{Sol}_p(A)$ such that, for every R-algebra B, the canonical function

$$\mathrm{Hom}_R(A, B) \longrightarrow \mathrm{Sol}_p(B), \qquad h \longmapsto h(a)$$

is an equivalence. Denote the type of this data by $\mathrm{Presents}(p, A)$, and put

$$\mathrm{FP}(A) = \left\| \sum_{p:\mathcal{P}_R} \mathrm{Presents}(p, A) \right\|.$$

This is a proposition in $\mathcal{U}_{\ell+1}$: its universal property quantifies over the algebras with small carriers.

Each p has a realisation

$$A_p = R[X_1, \ldots, X_n]/(f_1 - g_1, \ldots, f_m - g_m)$$

with its universal tuple. Two realisations of the same presentation have a unique isomorphism carrying one universal tuple to the other. Indeed, the two universal properties give maps in both directions; their composites fix the universal tuples and hence are identities.

Theorem 6.1. *There is a category* $\mathrm{Alg}_R^{\mathrm{fp}}$ *with objects and hom sets in* $\mathcal{U}_\ell$ *and a fully faithful inclusion into* Alg_R*. Its object type* I_R *satisfies*

$$I_R \simeq \sum_{A:\mathrm{Alg}_R} \mathrm{FP}(A). \tag{7}$$

If $a, b : I_R$ *have underlying algebras* A_a, A_b*, its hom set is* $\mathrm{Hom}_R(A_a, A_b)$*.*

Proof. Apply Theorem 3.1 to $p \mapsto A_p$. By (1), membership of A in the full image is equivalent to

$$\left\| \sum_{p:\mathcal{P}_R} \mathrm{Iso}_R(A_p, A) \right\|. \tag{8}$$

This proposition is equivalent to $\mathrm{FP}(A)$. A presentation of A gives an isomorphism $A_p \cong A$ by the common universal property. Conversely, an isomorphism $A_p \cong A$ gives an equality of algebras by Proposition 5.2; transporting the known realisation of p along it gives a realisation by A. Each use of a mere witness eliminates into a proposition.

For completeness, the object equivalence in (7) is explicit. Send a to (ia, u_a), where u_a is its finite-presentation proof. Conversely, membership gives a fibre point $(a, r : ia = A)$, so send (A, u) to a. The first round trip is the first coordinate of an equality in a fibre of i, which is a proposition. The second follows from r and the fact that $\mathrm{FP}(A)$ is a proposition. Fullness supplies the stated hom sets. □

The right side of (7) is expressed in $\mathcal{U}_{\ell+1}$, but the left side belongs to $\mathcal{U}_\ell$. Replacement has supplied a small model of the type of finitely presented algebras; no propositional resizing is used. The category contains all algebra homomorphisms between these objects.

For example, the presentation with no generators and no relations realises R, whereas the presentation with one generator X and relation $X = 0$ realises $R[X]/(X)$. Evaluation at zero gives

$$R[X]/(X) \cong R,$$

with inverse induced by the constants. These two codes therefore determine equal objects in I_R: the isomorphism gives a path by univalence of the small category. In general such paths retain the isomorphisms between algebras; in particular, loops at an object retain its automorphisms.

Let $\mathcal{C}_R$ be the precategory with finite presentations as objects and

$$\mathcal{C}_R(p, p') = \mathrm{Hom}_R(A_p, A_{p'}).$$

Its completion is $\mathrm{Alg}_R^{\mathrm{fp}}$, so Theorem 4.5 gives, for every category E,

$$\mathrm{Fun}(\mathrm{Alg}_R^{\mathrm{fp}}, E) \simeq \mathrm{Fun}(\mathcal{C}_R, E).$$

A functor described using presentation codes therefore descends to the algebras themselves, with uniqueness expressed by this equivalence. Different presentations are related through their algebra homomorphisms, rather than by a choice of one presentation for each algebra.

7. Set-valued functors and Yoneda

Let C be a small precategory: both its objects and its hom sets lie in $\mathcal{U}_\ell$. A covariant set-valued functor F assigns a small set $F(c)$ to each object and a function $F(f) : F(c) \to F(c')$ to each arrow, with

$$F(1_c)(x) = x, \qquad F(g \circ f)(x) = F(g)(F(f)(x)).$$

A natural transformation $\alpha : F \to G$ consists of functions $\alpha_c : F(c) \to G(c)$ satisfying

$$\alpha_{c'}(F(f)(x)) = G(f)(\alpha_c(x)).$$

Naturality is a proposition, so equality of transformations is pointwise equality of their components. For fixed F, G, the set $\mathrm{Nat}(F, G)$ lies in $\mathcal{U}_\ell$. The type of all such functors generally lies in $\mathcal{U}_{\ell+1}$, since its values range over small sets.

For $a : C$, the covariant representable functor is

$$h_a(c) = C(a, c), \qquad h_a(f)(u) = f \circ u.$$

Theorem 7.1 (Yoneda)**.** *For every a and every covariant set-valued functor F, evaluation at the identity is an equivalence*

$$\mathrm{Nat}(h_a, F) \simeq F(a), \qquad \alpha \longmapsto \alpha_a(1_a).$$

Its inverse sends $x \in F(a)$ to the transformation $u : a \to c \longmapsto F(u)(x)$.

Proof. The inverse formula is natural by the composition law for F. Evaluation of this transformation at 1_a gives $F(1_a)(x) = x$. Conversely, naturality of α at $u : a \to c$ gives

$$F(u)(\alpha_a(1_a)) = \alpha_c(u \circ 1_a) = \alpha_c(u).$$

Pointwise equality of transformations proves the second inverse law. □

Corollary 7.2 (Affine representables)**.** *On $C = \mathrm{Alg}_R^{\mathrm{fp}}$, let*

$$h_a(b) = \mathrm{Hom}_R(A_a, A_b).$$

Then

$$\mathrm{Nat}(h_a, h_b) \simeq \mathrm{Hom}_R(A_b, A_a).$$

The inverse sends $f : A_b \to A_a$ to precomposition with f: $u : A_a \to A_c \longmapsto u \circ f$.

Proof. Apply Theorem 7.1 to $F = h_b$. □

The representables thus give a fully faithful contravariant embedding

$$(\mathrm{Alg}_R^{\mathrm{fp}})^{\mathrm{op}} \longrightarrow [\mathrm{Alg}_R^{\mathrm{fp}}, \mathrm{Set}_\ell].$$

In detail, $y(1) = 1$, and for $f : b \to a$ and $g : c \to b$,

$$y(f \circ g) = y(g) \circ y(f).$$

These identities are respectively the unit law and associativity in the algebra category. This is the algebraic affine functor of points: a map out of the algebra A_a into a test algebra A_b is a point of h_a at A_b, and transformations between representables recover algebra maps in the opposite direction. All values, natural-transformation sets, and equivalences displayed here belong to $\mathcal{U}_\ell$. This size bound uses the small object type supplied by replacement.

Here both the represented algebras and the test algebras are finitely presented over R. Full faithfulness follows from Yoneda on exactly that category. A corresponding assertion for arbitrary schemes tested only on finitely presented algebras requires a separate argument and suitable hypotheses, or an enlarged test category; it does not follow from this application of Yoneda alone.

8. The categorical comparison for qcqs schemes

The affine Yoneda statement above uses finitely presented test algebras. For the scheme comparison in this section, tests are *all* commutative unital rings with set carriers in $\mathcal{U}_\ell$. For such a ring A, let $\operatorname{Spec} A$ be its locally ringed Zariski lattice, and put $h_A(B) = \mathrm{Hom}(A, B)$. The test object type lies in $\mathcal{U}_{\ell+1}$ and its hom sets lie in $\mathcal{U}_\ell$.

A geometric qcqs scheme is a small locally ringed lattice X merely admitting a finite affine cover. Its lattice and section-ring carriers lie in $\mathcal{U}_\ell$. The affine atlas is a property, rather than chosen structure; compact overlaps need not be affine. Let $\mathcal{G}_\ell$ be the precategory of these objects and their local geometric maps.

For small-set-valued functors on the test rings, the Zariski lattices form a functor Zar. A compact open of F is a natural transformation $U : F \to \mathrm{Zar}$, with realisation

$$|U|(B) = \sum_{x : F(B)} (U_B(x) = \top).$$

A functorial qcqs scheme is a Zariski sheaf merely admitting finitely many such opens U_i with pointwise join $\top$ and with each $|U_i|$ merely equivalent to some h_{A_i}. Sheafhood includes every finite basic cover, including the empty cover. Let $\mathcal{F}_\ell$ be the precategory of these functors and natural transformations. Its object condition, like geometric qcqs, is a proposition.

Proposition 8.1 (Univalence of both scheme categories)**.** *Both $\mathcal{G}_\ell$ and $\mathcal{F}_\ell$ are univalent categories.*

Proof. For commutative rings with set carriers, and for bounded distributive lattices, the structure identity argument of Proposition 5.2 applies: preservation of the operations under a carrier equivalence is the transport equation, and all laws are propositions.

Fix a lattice L. An identity of ring families over L corresponds to a family of ring isomorphisms. After identifying these families, naturality of the isomorphisms says precisely that the restriction maps agree. Function extensionality identifies the restriction families; their identity and composition laws are propositions. Sheafhood is also a proposition. For a fixed presheaf, the invertibility open of a section is unique by its defining universal property, so the type of

invertibility maps is a proposition. Their support laws are propositions too. Thus the locally ringed conditions add no further identity data.

A geometric isomorphism $f : X \to Y$ gives a lattice isomorphism $f^* : L_Y \to L_X$ and compatible ring isomorphisms

$$f_u^\sharp : \mathcal{O}_Y(u) \longrightarrow \mathcal{O}_X(f^*u).$$

Conversely such data give an isomorphism; it and its inverse are local because ring isomorphisms preserve and reflect units, hence preserve the uniquely characterised invertibility opens. Apply the lattice identity principle, retaining the inverse-image direction, then the fixed-lattice ring-family argument above. These give contractible total spaces of isomorphisms of locally ringed lattices. Passing between structural isomorphisms and actual geometric isomorphisms preserves the whole ringed map. Lemma 1.2 therefore proves univalence for the category of all these spaces.

For functors, an isomorphism is a natural transformation whose components are equivalences. Pointwise univalence and function extensionality identify its value families. After that identification, naturality is equality of the reindexing maps; sethood and the functor laws are propositions. Lemma 1.2 proves univalence of the category of small-set-valued functors.

Finally, a full subcategory defined by a propositional object condition inherits univalence. Its isomorphisms are the original ones, and proofs of the object condition add no identity data. Apply this to the two qcqs conditions to obtain the assertion. □

The geometric category has objects in $\mathcal{U}_{\ell+1}$ and hom sets in $\mathcal{U}_\ell$. For the functorial category both bounds are initially $\mathcal{U}_{\ell+1}$. We may therefore place both categories at the common bound $\ell + 1$; the comparison uses no resizing of natural-transformation types.

Theorem 8.2 (Equivalence of qcqs scheme categories)**.** *The functor of points*

$$H : \mathcal{G}_\ell \longrightarrow \mathcal{F}_\ell, \qquad h_X(B) = \mathrm{Hom}(\mathrm{Spec}\, B, X), \qquad H(f)_B(p) = f \circ p,$$

is an equivalence of univalent categories. In particular,

$$\mathrm{Ob}(\mathcal{G}_\ell) \simeq \mathrm{Ob}(\mathcal{F}_\ell).$$

It has an explicit inverse functor and natural isomorphisms for both composites, without an axiom of choice.

Proof. The geometric sheaf and atlas theorems make h_X a functorial qcqs scheme. The geometric unit and associativity laws make H a functor. The checked absolute Hom–Nat theorem gives full faithfulness:

$$\mathrm{Hom}(X, Y) \simeq \mathrm{Nat}(h_X, h_Y).$$

The checked absolute essential-image theorem gives, for each $F \in \mathcal{F}_\ell$,

$$\left\| \sum_{X : \mathcal{G}_\ell} (F \simeq h_X) \right\|.$$

Here the displayed equivalences are natural. Inverting them gives mere essential surjectivity of H.

For clarity, this input reconstructs an actual finite atlas using geometric transition isomorphisms on compact overlaps. Their triple domains are identified before the cocycle equations are compared. Finite gluing gives a small geometric qcqs space, and compatibility in both directions identifies its functor with F. The merely given atlas and affine witnesses are eliminated only into mere representability. The companion note, *The essential image of the functor of points*, gives this construction and its detailed proof.

Proposition 8.1 supplies univalence of both categories. Theorem 4.4 now applies to H. In particular it makes each representation fibre a proposition before removing the truncation. Formula (5) gives the inverse on arrows, with its functor laws and both natural isomorphisms. □

The theorem retains automorphisms as loops in the object types. It compares geometric schemes with functorial schemes, rather than passing to their sets of isomorphism classes. The theorem also has the following relative form, obtained by totalising the native functor on algebra tests.

Theorem 8.3 (Relative qcqs comparison)**.** *Let k be a small commutative unital ring with set carrier. Let $\mathcal{F}_{k,\ell}$ be the category of native small-set-valued functors on all small k-algebras satisfying the relative Zariski sheaf and finite affine compact-open atlas conditions. The relative functor of points*

$$H_k : \mathcal{G}_\ell/\operatorname{Spec} k \longrightarrow \mathcal{F}_{k,\ell}, \qquad H_k(X,s)(B,\iota) = \operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec}(B,\iota),(X,s)),$$

is an equivalence of univalent categories, with an explicit inverse functor and both natural isomorphisms.

Proof. For a native relative functor F, define its totalisation by

$$(\mathsf{T}F)(B) = \sum_{\iota : k \to B} F(B,\iota).$$

On $h : B \to C$, reindex (ι, x) along the canonical algebra map $c_{h,\iota} : (B,\iota) \to (C, h\iota)$. If $p : h\iota = \kappa$, path induction gives the precise dependent equation

$$\operatorname{transport}_{j \mapsto F(C,j)}(p, F(c_{h,\iota})(x)) = F(h,p)(x).$$

Thus total reindexing on any actual algebra map a identifies the whole point with $(\kappa, F(a)(x))$. This proves the total functor laws and naturality of its projection $\pi_F : \mathsf{T}F \to h_k$.

Intrinsic relative sheafhood is equivalent to absolute sheafhood of $\mathsf{T}F$. For the forward direction, project a compatible total cover family (ι_i, x_i) to base maps and glue them in the sheaf h_k. Rebase each x_i along the resulting equation $\lambda_i \iota = \iota_i$ before applying the relative sheaf condition. The whole-pair reindexing equation gives compatibility on the fixed base, and base uniqueness followed by relative uniqueness proves total uniqueness. Conversely, total gluing of paired relative values and base separation in h_k give a value in the specified fibre. Injectivity of pairing at that fixed base proves relative uniqueness. Empty covers are included in both arguments.

Compact opens correspond by evaluation at (ι, x), and realisations commute with totalisation by reordering the membership proof. Totalising a relative affine representable gives h_A: its ring map $h : A \to B$ forces the target base map to be $h\alpha$, where $\alpha : k \to A$ is the source structure. Conversely, an absolute affine chart gives α by projecting its generic point. Naturality supplies the base path needed to transport its values into each specified relative fibre. Thus relative and total affine atlas conditions agree. Together, these prove

$$F \text{ is relative functorially qcqs} \quad \Longleftrightarrow \quad \mathsf{T}F \text{ is absolute functorially qcqs.}$$

Apply the absolute essential-image theorem to $\mathsf{T}F$. For an actual representing equivalence $e : \mathsf{T}F \simeq h_X$, the natural projection $\pi_F e^{-1} : h_X \to h_k$ gives $s : X \to \operatorname{Spec} k$ by the affine comparison and full faithfulness. Its equation on affine probes identifies the projection fibres with points over k. On the original side the fibre over ι is

$$\sum_{\kappa : k \to B} \sum_{x : F(B,\kappa)} (\kappa = \iota) \simeq F(B,\iota),$$

by transport along the displayed path. Its inverse pairs with ι and reflexivity; path induction proves both inverse laws. Restricting e to these fibres gives a native relative representing equivalence. Only mere relative representability is asserted before eliminating the mere absolute representation. Conversely total relative geometric points are h_X, since each ordinary point has a unique base lift. The totalisation criterion gives the relative geometric qcqs property. This proves mere essential surjectivity of H_k. The existing relative Hom–Nat theorem gives full faithfulness on the same native functors.

For univalence, the geometric category is the slice of a univalent category. Its isomorphisms are base isomorphisms $e : x \cong y$ satisfying $te = s$; their inverses commute since $se^{-1} = t$. Over the identity base isomorphism, the compatible structure arrows form the contractible singleton $(t, t1 = s)$. The total-space criterion therefore proves slice univalence. On the functorial side, the value-family and action identity argument applies to the fixed type of actual algebra tests, and the relative qcqs condition is a proposition. Both categories are univalent. Theorem 4.4 now supplies the stated equivalence and its explicit quasi-inverse. □

The detailed transport, rebasing, and fibre arguments, with their correspondence to the checked declarations, are given in the companion note *Relative totalisation over a base ring*. The native sheaf and atlas conditions are defined on algebra tests before the intrinsic-total equivalence is proved; they are not abbreviations for absolute conditions on $\mathsf{T}F$.

9. Connection with the implementation

The preceding constructions are implemented in Unit, in `sag/categories/`, `sag/fpalg/`, and `sag/qcqs/`. The categorical foundations check with 21 modules, and the finitely presented algebra development with 47 modules, including dependencies. The complete absolute and relative qcqs comparison checks with 249 modules and 9,250 declarations. All three checkpoints have zero failures. The informal arguments are recorded in the design documents and at the heads of the source files. The correspondence is as follows.

Mathematical construction	**Source files**
Categories and the total-space criterion	`Category.seg`
Functors and natural transformations	`Functor.seg`, `FunctorCat.seg`
Small full image (Theorem 3.1)	`SmallImage.seg`
Full faithfulness and descent	`PrecomposeFF.seg`, `Descent.seg`, `UniversalProperty.seg`
Univalence and the object equivalence	`FunctorUnivalence.seg`, `EquivObjects.seg`
Completion property (6)	`CompletionUP.seg`
Algebra category and structure identity	`AlgebraCategory.seg`, `AlgebraSIP.seg`
Small finitely presented algebra category	`FPImage.seg`, `FPComparison.seg`
Descent from presentation codes	`FPDescent.seg`
Yoneda and affine representables	`Yoneda.seg`, `FPYoneda.seg`
Equivalence from mere essential surjectivity	`WeakCategoryEquivalence.seg`, `EmbeddingFibres.seg`
Explicit inverse and natural isomorphisms	`CategoryQuasiInverse.seg`
Univalent geometric and functorial categories	`GeometricCategory.seg`, `FunctorCategory.seg`, `FullSubcategory.seg`
Qcqs comparison (Theorem 8.2)	`QcqsCategories.seg`

The category interface is `Category[k,l]`: objects have level k and hom sets level l. For a family indexed at level l in such a category, `SmallImage.SmallFullImage` gives a `Category[l,l]`. Its image fibre is a proposition, and `from-fibre` and `to-fibre` express (1). The completion theorem exposes both full faithfulness and an equivalence on functor types. Functor categories are placed in a common universe bounding the source and target levels.

For algebras, `AlgebraSIP.Over.category` has levels $l+1, l$. The algebra representation and presentation universal properties are reused from `sag/presentations/`; the structure identity proof uses `SipGeneric` and local structure pieces for negation and the map from the base ring. `FPImage.Over.SFI.category` has levels l, l. The statements `FPComparison.Over.comparison` and `FPComparison.Over.objects` give respectively membership $\simeq \mathrm{FP}(A)$ and (7). Thus the comparison is with the existing universal-property definition of finite presentation, not merely with a new predicate on codes.

Set-valued functors are represented directly by their values, functions on arrows, and pointwise laws. Natural transformations are component families with pointwise naturality; `nat-eq` proves their extensionality. This interface suffices for `Yoneda.yoneda`, and `FPYoneda` states the instance in algebra-homomorphism notation. The precomposition map is proved to be an equivalence, and its identity and reversed composition laws are checked. No packaging of all these set-valued functors as a univalent category of functors into a category of sets is needed for these results.

The foundational parameters are universe univalence and the two parts of replacement: small images and their surjectivity elimination principle. Function extensionality and propositional univalence are derived. The small full image uses replacement once; the algebra application

uses it at levels $l, l+1$. The object equivalence does not use propositional resizing. Later Yoneda arguments use the small category already constructed and require no further completion.
For the scheme comparison, the sources in `sag/qcqs/` reuse `SipGeneric` and the existing category interfaces through local symlinks, without modifying their implementations. `RingIdentity` and `LatticeIdentity` prove the algebraic identity principles. `FamilyIdentity` and `DecoratedIdentity` pass them to indexed families, structure, and propositional properties. `PresheafIdentity` and `LocalIdentity` handle restrictions, sheafhood, and locality. `SpaceIdentity` and `SpaceIsoIdentity` connect the structural presentation to the actual geometric maps, with an equation of their underlying ringed maps. The total-space criterion then gives `GeometricCategory.univalent`. `SetIdentity` and `FunctorIdentity` supply the corresponding argument for `FunctorCategory.univalent`.
`QcqsCategories.functor` is the existing functor of points; `fully-faithful` uses `PointsUniversal`, and `essentially-surjective` uses `EssentialImage.reconstruct`. The source and target are actual univalent category records, obtained by `FullSubcategory`. Their univalence proofs are constructed, rather than additional parameters of the comparison.
`QcqsCategories.object-equivalence` records the equivalence of object types.
`QcqsCategories.equivalence` packages the explicit inverse functor, its identity and composition laws, and the two natural isomorphisms, including both naturality proofs.
For Theorem 8.3, `RelativeTotal.change-target` states the displayed transport comparison and `on-algebra` identifies the whole total point on an actual algebra map. `RelativeBaseGlue`, `RelativeTotalSheaf`, and `RelativeSheafOfTotal` implement base gluing, rebasing and the two sheaf implications. `RelativeTotalisation` exposes the sheaf and qcqs equivalences; `RelativeAffineOfTotal` recovers the generic algebra structure. `RelativeFibreRecovery`, `RelativeFibreEquivalence`, and `RelativeGeometricFibre` prove the fibre identifications, retaining the dependent base equations. `RelativeEssentialImage` proves both directions of relative representation.
`SliceCategory` proves univalence of the geometric slice, and `RelativeFunctorIdentity` with `RelativeFunctorCategory` proves native functor-category univalence. The final statement is `RelativeCategories.Over[K].equivalence`, with the inverse functor and both natural isomorphisms supplied by the same general `CategoryQuasiInverse` used absolutely.
The full independent qcqs check has 9,250 declarations and zero failures. The combined `essential-image.opens`, `categories.opens`, and `relative.opens` audit has 3,752 opening lists; every reported list says that nothing else opened. The absolute categorical continuation has 591 declarations in 18 modules and reuses 167 declarations in five source modules. The relative continuation adds 863 audited declarations in 35 modules. No core or category refactoring, new axiom, choice principle, or resizing principle is used.
Thus this checkpoint includes the categorical foundations and both the absolute and relative equivalences of geometric and functorial qcqs schemes.